A project report on

IOT-BASED BLOCKCHAIN FRAMEWORK FOR ORGANIC FOOD TRACEABILITY

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology in Computer Science and Engineering

by

GOVARDHINI POLISETTI (22BCE8163)

SRIVALLI GABBURI (22BCE8054)

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

May, 2026

DECLARATION

I here by declare that the thesis entitled "**IoT-based Blockchain Framework for Organic Food Traceability**" submitted by me, for the award of the degree of Bachelor of Technology in Computer Science, VIT-AP, is a record of bonafide work carried out by me under the supervision of **Dr. Shalini Subramani.**

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

G.Srivalli

Place: Amaravati P.Govardhini

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ABSTRACT

The increasing demand for organic food has created a strong need for reliable systems that can genuinely verify whether a product is organic rather than depending only on labels or manual certification processes. In traditional approaches, verification is mainly based on self-declared data, periodic inspections, and paper-based records, which are not continuous, can be inaccurate, and are often vulnerable to manipulation, thereby reducing consumer trust. To overcome these limitations, this project proposes a practical and automated organic food traceability framework that integrates Internet of Things (IoT), edge computing, and blockchain technology to ensure transparency and authenticity throughout the food lifecycle. In this system, IoT sensors deployed on farms continuously collect real-time data such as soil moisture, temperature, humidity, and indicators of chemical usage. Instead of directly storing raw sensor data, an edge verification layer processes and validates the data by filtering anomalies, checking for abnormal patterns, and applying predefined organic compliance rules to ensure contextual correctness. Only verified and meaningful data is then transmitted to the blockchain, where it is immutably stored using smart contracts, ensuring that records cannot be altered or tampered with. The system further introduces a rule-based compliance mechanism that automatically evaluates whether farm conditions satisfy organic standards, along with a dynamic trust scoring model that quantifies the reliability of farmers over time based on their historical data and consistency of compliance. A complete working prototype has been developed, consisting of a blockchain layer, backend server, edge simulation module, and role-based web interfaces for farmers, customers, and administrators. Farmers can monitor sensor readings and track their trust score, customers can verify product authenticity through QR-based traceability, and administrators can oversee system performance and detect violations without directly modifying blockchain data. By reducing dependence on manual verification, enabling continuous real-time monitoring, and ensuring secure and transparent data storage, the proposed system significantly improves data integrity and trust across the supply chain. Overall, this approach provides a scalable, automated, and reliable solution for organic food traceability and certification from farm to consumer.

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Place: Amaravati — Srivalli Gabburi,Govardhini Polisetti

Date: May 8th, 2026 — **Name of the student**

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LIST OF ABBREVIATIONS

ABREVIATIONS	-	FULL FORMS
IOT	-	INTERNET OF THINGS
MQTT	-	MESSAGE QUEUING TELEMETRY TRANSPORT
LoRaWAN	-	LONG RANGE WIDE AREA NETWORK
GPS	-	GLOBAL POSITIONING SYSTEM
NPOP	-	NATIONAL PROGRAMME FOR ORGANIC PRODUCTION
SC	-	SMART CONTRACT
API	-	APPLICATION PROGRAMMING INTERFACE
QR	-	QUICK RESPONSE
RBAC	-	ROLE-BASED ACCESS CONTROL
DLT	-	DISTRIBUTED LEDGER TECHNOLOGY
GW	-	GATEWAY
UI	-	USER INTERFACE
CPU	-	CENTRAL PROCESSING UNIT
JSON	-	JAVASCRIPT OBJECT NOTATION
REST	-	REPRESENTATION STATE TRANSFER

Chapter 1
INTRODUCTION

1.1 BACKGROUND

The demand for organic food has grown rapidly across the world in recent years, mainly due to increasing awareness among consumers about health, environmental sustainability, and food safety. People today are more conscious about what they eat and prefer food that is free from harmful chemicals, pesticides, and synthetic fertilizers. Organic farming practices are designed to maintain soil health, protect biodiversity, and reduce environmental impact. Because of this, organic products are often considered safer and more environmentally friendly compared to conventional food products. However, as the demand for organic food increases, so does the challenge of ensuring that these products are genuinely organic and not falsely labeled.

At present, most organic certification systems rely heavily on manual processes such as paper-based records, self-declared information provided by farmers, and periodic inspections carried out by regulatory authorities. These inspections usually happen at fixed intervals, meaning they only capture the condition of the farm at a specific point in time rather than continuously monitoring it. This creates gaps in verification, allowing farmers or intermediaries to manipulate records or temporarily comply with standards only during inspection periods. As a result, it becomes difficult to detect violations in real time, and consumers may end up trusting labels that do not truly reflect the actual farming practices. This lack of continuous monitoring and dependence on human intervention reduces transparency and weakens trust across the supply chain.

To address these issues, researchers and industries have explored the use of blockchain technology for improving food traceability. Blockchain provides a decentralized and tamper-proof system where data, once recorded, cannot be easily altered. This makes it highly useful for maintaining transparent records of food movement from farm to consumer. However, while blockchain ensures that stored data is secure and immutable, it does not

guarantee that the data being entered is accurate or truthful. Most blockchain-based traceability systems still depend on manually entered information, which means that if incorrect or false data is uploaded, the system will preserve that incorrect information permanently.

On the other hand, Internet of Things (IoT) technologies have been widely adopted in smart agriculture to monitor real-time conditions such as soil moisture, temperature, humidity, and other environmental parameters. These systems provide continuous and automated data collection directly from the farm, reducing the need for manual input and improving data accuracy. However, IoT systems alone lack secure storage mechanisms and do not provide strong guarantees against data tampering or unauthorized modification. They also do not include automated mechanisms to verify whether the collected data meets organic compliance standards.

Therefore, there is a clear need for an integrated approach that combines the strengths of both IoT and blockchain technologies. By merging real-time data collection with secure, tamper-proof storage and automated verification, it is possible to create a more reliable and transparent system for organic food traceability. This forms the foundation for developing advanced solutions that can ensure continuous monitoring, reduce human intervention, and build greater trust between farmers, regulators, and consumers.

1.2 PROBLEM STATEMENT

Even though technologies like IoT and blockchain have developed significantly in recent years, there is still a major gap when it comes to combining these technologies into a single, reliable system for organic food traceability. Most existing systems either focus only on collecting data or only on storing data securely, but very few ensure that the data itself is trustworthy. In many cases, data is manually entered without proper verification, which means that incorrect or false information can still be recorded and treated as valid. Additionally, current systems do not provide sufficient context about farming activities, such as why a particular action was taken, under what conditions it occurred, or whether it

aligns with organic standards. This lack of context makes it difficult to fully understand and validate farming practices.

Another major limitation is the absence of continuous monitoring throughout the entire lifecycle of food production. Most systems only track specific stages, such as production or distribution, instead of providing complete visibility from seed to shelf. Because of this, important events or violations may go unnoticed. Furthermore, existing solutions do not include mechanisms to dynamically evaluate the credibility of producers over time. Trust is often assumed rather than measured, which can lead to misuse of organic labels and reduced consumer confidence. To overcome these challenges, this research proposes an integrated IoT-based blockchain framework that combines real-time data collection, intelligent validation, secure storage, and automated compliance checking to create a more transparent and trustworthy organic food traceability system.

1.3 OBJECTIVES

The primary objective of this project is to design and develop a comprehensive system that ensures reliable and transparent organic food traceability by integrating multiple modern technologies. The first objective is to enable real-time data collection from farms using IoT sensors. These sensors continuously monitor important environmental and soil parameters such as temperature, humidity, soil moisture, and chemical usage indicators. By capturing data directly from the farm environment, the system reduces dependency on manual inputs and improves the accuracy and authenticity of the collected information.

The second objective is to implement an edge-based validation mechanism that processes the collected data before it is stored. This layer plays a critical role in ensuring data quality by filtering out abnormal or inconsistent readings, validating sensor outputs, and enriching the data with contextual details such as time, location, and farming conditions. This step ensures that only meaningful and trustworthy data enters the system, thereby increasing reliability.

The third objective is to utilize blockchain technology for secure and immutable data storage. Blockchain acts as a decentralized ledger where validated data is recorded

permanently, making it resistant to tampering or unauthorized modification. This ensures transparency and builds trust among all stakeholders involved in the supply chain.

The fourth objective is to implement smart contracts that automatically evaluate whether the recorded data complies with organic farming standards. These contracts enforce predefined rules and conditions, allowing the system to detect violations and ensure that only compliant practices are maintained without requiring manual intervention.

Finally, the project aims to develop a dynamic trust scoring model that evaluates the credibility of farmers over time. This model analyzes historical data and compliance behavior to assign a trust score, which helps in identifying reliable producers and detecting potential risks. Together, these objectives aim to create a system that is automated, transparent, and capable of ensuring true organic verification.

1.4 SCOPE OF THE PROJECT

The scope of the AgriTrace system is designed to cover the complete lifecycle of organic food production, starting from cultivation at the farm level and extending up to the final purchase by the consumer. The system integrates multiple layers to ensure end-to-end traceability and transparency. At the IoT layer, real-time data is collected using sensors that monitor environmental and soil conditions. This ensures continuous observation of farming activities without relying on manual data entry.

The edge layer is responsible for validating and processing the collected data before it is stored. It performs checks for data accuracy, removes inconsistencies, and adds contextual information such as location, timestamp, and environmental conditions. This ensures that the data entering the system is reliable and meaningful.

The blockchain layer serves as a permanent and secure storage mechanism where validated data is recorded. Once stored, the data cannot be altered, ensuring integrity and transparency across the supply chain. Smart contracts within this layer automate compliance verification by evaluating whether the farming practices meet organic standards.

At the application layer, user-friendly dashboards are provided for different stakeholders. Farmers can monitor their farm conditions and trust scores, customers can verify product authenticity through traceability information (such as QR codes), and administrators can oversee system performance and detect any violations. Importantly, these dashboards provide read-only access, ensuring that no stakeholder can manipulate stored data.

The key contributions of this system include sensor-based verification instead of manual input, context-aware data validation, continuous lifecycle monitoring, automated compliance enforcement, and a dynamic trust scoring mechanism. Although the current implementation uses simulated IoT data for testing and demonstration purposes, the system is designed to support real-world sensor deployment in future work. Overall, the project aims to provide a scalable, transparent, and trustworthy solution for next-generation organic food traceability, supporting multiple stakeholders including farmers, customers, and administrators.

1.5 ORGANIZATION OF THE REPORT

Chapter 2 reviews the existing literature on IoT-based smart agriculture, blockchain-based food traceability, and hybrid IoT-blockchain frameworks.

Chapter 3 presents the detailed system architecture and methodology of AgriTrace.

Chapter 4 provides results and analysis based on simulated experiments.

Chapter 5 discusses findings, implications, and limitations.

Chapter 6 concludes the report and outlines directions for future work.

Chapter 2

REVIEW OF LITERATURE

2.1 IOT-BASED SMART AGRICULTURE

IoT-based smart agriculture systems have become an important area of research due to their ability to continuously monitor farming conditions and improve productivity. These systems use sensors and connected devices to collect real-time data such as soil moisture, temperature, humidity, and crop health, which helps farmers make better decisions. Although IoT has greatly improved data collection and automation in agriculture, most existing solutions focus only on monitoring and do not address secure storage, data verification, or compliance with organic farming standards.

Kour and Arora (2020) presented a detailed survey on the use of IoT in agriculture, focusing on how technology can support precision farming practices. Their study highlights the role of sensors, wireless communication, and automated systems in improving crop yield and resource management. They discuss applications such as automated irrigation systems, where water is supplied based on soil moisture levels, and remote monitoring systems that allow farmers to track farm conditions from anywhere. While their work demonstrates the effectiveness of IoT in increasing efficiency and reducing manual effort, it mainly focuses on data collection and automation, without addressing issues related to data security, authenticity, or long-term traceability.

Ayaz et al. (2019) explored different architectures of IoT-based smart agriculture systems and explained how these systems can be used for large-scale field monitoring. Their research emphasizes the use of distributed sensor networks and cloud-based platforms to gather and analyze agricultural data. They also discuss how IoT can help in improving decision-making by providing real-time insights into environmental conditions. However, their approach relies heavily on centralized systems, which can be vulnerable to data manipulation or unauthorized access. Additionally, their work does not include

mechanisms for verifying whether the collected data is accurate or for ensuring that it meets specific standards such as organic compliance.

Farooq et al. (2022) proposed an IoT-based smart greenhouse system designed to maintain optimal environmental conditions for crop growth. Their framework includes sensors and control systems that automatically adjust temperature, humidity, and lighting within the greenhouse to create ideal conditions for plants. This approach improves crop quality and reduces human intervention by automating environmental control. While this system is effective for controlled farming environments, it is mainly focused on optimization and efficiency rather than transparency or traceability. It does not address how the collected data can be securely stored, verified, or used to prove compliance with organic farming practices.

Alzubi and Galyna (2023) examined the integration of Artificial Intelligence (AI) with IoT technologies to support sustainable agriculture. Their research highlights how AI can analyze large volumes of sensor data to provide predictive insights, such as identifying crop diseases, optimizing resource usage, and improving yield. The combination of AI and IoT enables smarter and more efficient farming practices. However, despite these advancements, their work does not consider secure data storage or tamper-proof mechanisms. There is also no focus on traceability or verification of organic standards, which limits its application in building trust among consumers.

2.2 BLOCK-CHAIN BASED FOOD TRACEABILITY

Blockchain technology has gained significant attention in recent years as a powerful solution for improving transparency, traceability, and trust in agri-food supply chains. Its decentralized and immutable nature ensures that once data is recorded, it cannot be easily altered or deleted, making it suitable for maintaining reliable records of food movement from farm to consumer. Many researchers have explored blockchain for tracking products across different stages such as production, processing, distribution, and retail. However, despite its advantages in secure storage, most blockchain-based systems still depend on

manually entered data, which limits their ability to guarantee that the information being stored is accurate and trustworthy.

Tripoli and Schmidhuber (2018) highlighted the potential of blockchain technology in transforming global food systems by improving transparency and reducing fraud. Their work explains how blockchain can create a shared and immutable ledger that records every transaction across the supply chain, making it easier to track the origin and movement of food products. They emphasize that such transparency can increase consumer trust and reduce the chances of data manipulation by intermediaries. However, their study mainly focuses on the benefits of secure record-keeping and does not address how the data being entered into the blockchain can be verified for accuracy. As a result, if incorrect or false data is recorded, blockchain will still preserve it without validation.

Ellahi et al. (2023) provided a comprehensive review of various blockchain-based traceability frameworks used in food supply chains. Their research analyzes how different systems use blockchain to ensure data integrity, improve traceability, and enhance coordination among stakeholders. They discuss multiple architectures and highlight the advantages of decentralization and transparency in maintaining trustworthy records. While their work demonstrates that blockchain can effectively protect stored data from tampering, it also points out that many systems still rely on external data sources that are not always verified. This means that although the data is secure, its authenticity is not guaranteed, which remains a major limitation.

Zhao, Sahu, and Wang (2020) studied the role of blockchain in managing agri-food value chains and improving supply chain efficiency. Their research focuses on how blockchain can streamline operations, reduce delays, and enhance collaboration among different participants such as farmers, distributors, and retailers. They explain that blockchain can help in tracking product movement and ensuring accountability at each stage. However, their approach primarily deals with supply chain management and does not focus on farm-level data collection or verification. The system still depends on participants to input data manually, which introduces the risk of inaccurate or manipulated information.

Lin, Wang, and Li (2022) proposed an integrated approach that combines IoT and blockchain technologies for food traceability. Their work attempts to bridge the gap between real-time data collection and secure storage by using IoT devices to gather environmental data and blockchain to store it. This integration improves transparency and provides better visibility into the food production process. However, their framework does not include a strong validation mechanism before storing data on the blockchain. Without proper filtering or compliance checks, there is still a possibility of incorrect or noisy sensor data being recorded. Additionally, their system does not include automated mechanisms for evaluating whether the data meets organic farming standards.

2.3 HYBRID IOT-BLOCKCHAIN TRACEABILITY

To overcome the limitations of using IoT or blockchain individually, several researchers have proposed hybrid frameworks that combine real-time data collection with secure data storage. These systems aim to use IoT for continuous monitoring and blockchain for tamper-proof record keeping. In addition, some approaches integrate Artificial Intelligence (AI) to enhance decision-making and predictive capabilities. While these hybrid systems show significant improvements in automation, data security, and system intelligence, most of them do not specifically focus on organic food traceability, continuous compliance verification, or providing contextual understanding of farming practices.

Kaushik, Jain, and Prakash (2025) proposed an advanced framework that combines Artificial Intelligence and blockchain for secure agricultural data management and crop prediction. Their system uses IoT devices to collect farm data and AI models to analyze this data for predicting crop yield and improving farming decisions. Blockchain is used to securely store the processed data and ensure that it cannot be tampered with. This integrated approach enhances both decision-making and data security. However, the primary focus of their work is on improving agricultural productivity and prediction accuracy rather than verifying organic farming practices. The system does not include mechanisms to enforce organic compliance rules or to evaluate whether the farming methods meet certification standards.

Elbasi et al. (2023) conducted a comprehensive review of the use of Artificial Intelligence in agriculture, highlighting how AI can improve efficiency, reduce resource usage, and support sustainable farming. Their work discusses applications such as disease detection, crop monitoring, and predictive analytics using data collected from IoT devices. In their later work, Elbasi et al. (2025) introduced the concept of green AI models, which focus on developing energy-efficient and environmentally sustainable solutions for smart agriculture. These models aim to reduce computational cost while maintaining high performance in predictive tasks. Although both studies emphasize sustainability and intelligent data analysis, they do not address secure data storage, traceability, or verification of organic compliance using blockchain, which limits their application in building trust for organic food systems.

A recent study on urban horticulture presented at ICSD (2024) introduced a networked-intelligence blockchain framework designed to support organic food production in urban environments. This system integrates IoT devices, data analytics, and blockchain to create a connected ecosystem for monitoring and managing urban farming activities. The framework improves data sharing, transparency, and coordination among stakeholders. However, while it supports traceability and data integration, it does not provide strong mechanisms for validating the authenticity of the collected data before storing it on the blockchain. Additionally, it lacks continuous compliance enforcement and does not include a dynamic trust scoring model to evaluate the reliability of producers over time.

2.4 RESEARCH GAPS

A detailed review of existing literature on IoT-based agriculture, blockchain-based traceability, and hybrid IoT–blockchain frameworks reveals several important limitations that still remain unaddressed. While significant progress has been made in improving data collection, storage, and transparency, most current systems focus only on specific aspects rather than providing a complete and reliable solution for organic food traceability. These gaps highlight the need for a more integrated and intelligent framework that can ensure both data authenticity and continuous verification.

One of the major gaps identified is the lack of proper integration between sensor-verified data and blockchain systems for real-time organic compliance enforcement. Although IoT systems are capable of collecting real-time environmental data from farms, this data is often not directly linked with blockchain-based verification mechanisms. In many cases, blockchain systems still depend on manually entered or externally provided data, which reduces their effectiveness in ensuring true authenticity. Without direct sensor integration and automated validation, the system cannot guarantee that the data stored on the blockchain accurately represents actual farm conditions.

Another important limitation is that most existing solutions focus primarily on supply chain traceability rather than covering the entire lifecycle of food production. These systems typically track the movement of products from farm to market but do not continuously monitor farming activities from the cultivation stage itself. As a result, critical information about how the food was grown, whether organic standards were followed throughout, and whether any violations occurred during production is often missing. This lack of full lifecycle monitoring reduces the reliability of traceability systems and limits their ability to provide complete transparency.

The absence of context-aware validation is another significant research gap. Current systems generally record raw data without providing meaningful context about farming actions. For example, while a system may record temperature or soil moisture values, it does not explain why certain conditions occurred, whether they are normal for that region or crop type, or how they affect organic compliance. Without contextual understanding, it becomes difficult to interpret data accurately and make informed decisions about whether farming practices meet required standards.

In addition, most existing frameworks do not include a dynamic trust scoring mechanism to evaluate the credibility of producers over time. Trust is usually assumed based on certifications or historical reputation, rather than being continuously measured using real-time data. This creates a limitation in identifying unreliable producers or detecting patterns of non-compliance.

Chapter 3

SYSTEM ARCHITECTURE AND METHODOLOGY

This chapter presents the detailed architecture, design decisions, and methodology of the AgriTrace framework. The system follows a multi-layered approach that integrates IoT sensing, edge-based validation, blockchain storage, and smart contract-driven compliance evaluation to ensure that only verified, contextually enriched data is permanently recorded.

3.1 SYSTEM ARCHITECTURE OVERVIEW

The AgriTrace system is designed as a four-layer architecture, each layer serving a distinct and clearly separated responsibility. The layered design ensures modularity, scalability, and efficient data processing. The four layers are:

- **Layer 1 –** IoT Sensing Layer: Physical field hardware that continuously monitors organic farm conditions.
- **Layer 2** – Edge Verification Layer: Local validation and context enrichment before data reaches the chain.
- **Layer 3** – Blockchain Layer: Immutable trust anchor; no record can be altered after commit.
- **Layer 4** – Application Layer: Role-gated web dashboards; no user can write to the ledger.

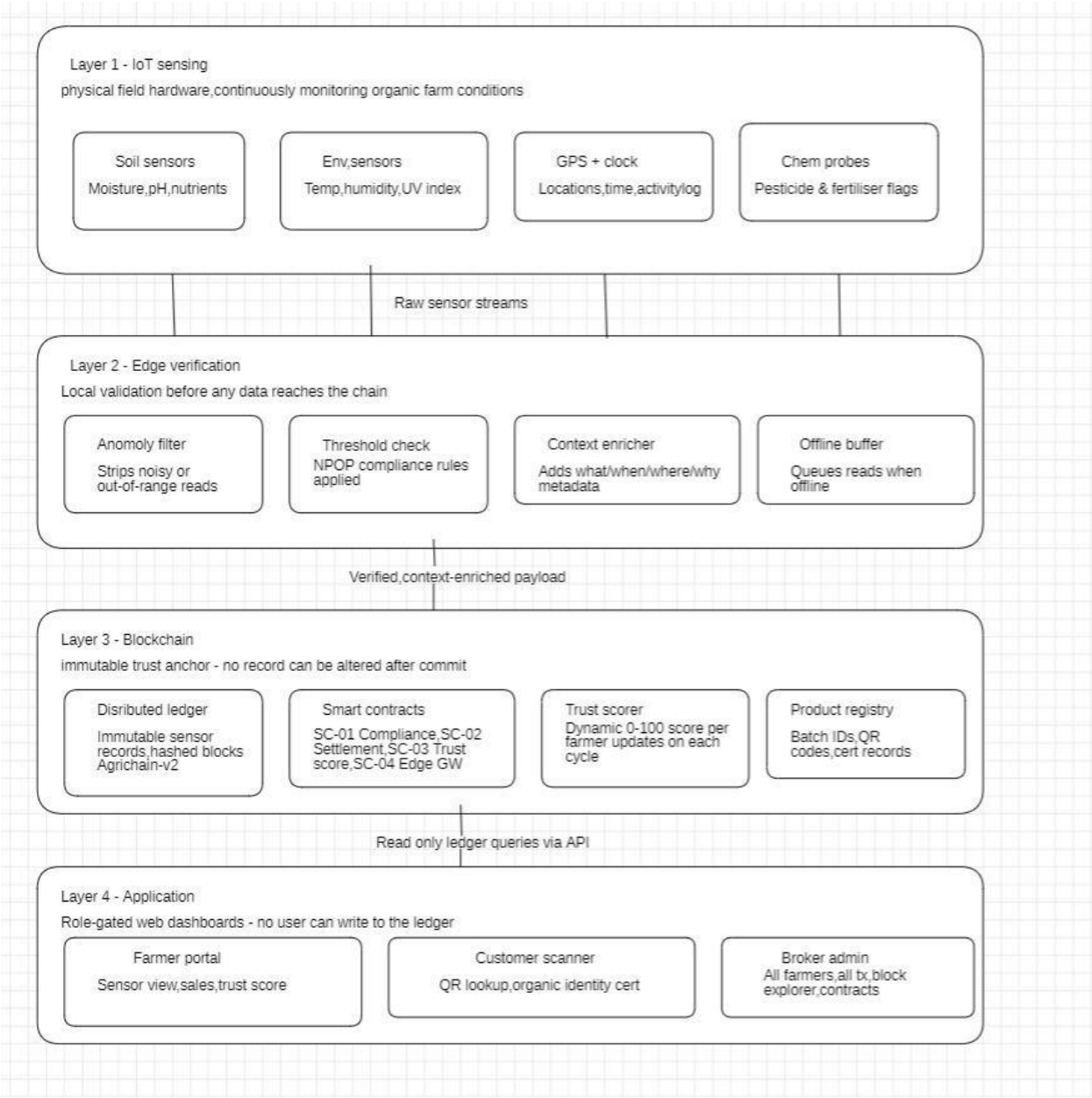

Fig. 3.1. Layered architecture of the AgriTrace IoT–Blockchain traceability system.

3.2 IOT SENSING LAYER

The IoT sensing layer forms the foundation of the AgriTrace system, as it represents the actual farm environment where data is generated. This layer consists of physical sensors and devices deployed across organic farms that continuously monitor real-world conditions.

Unlike traditional systems that depend on manual data entry, this layer ensures that information is collected automatically and in real time. By directly capturing data from the field, the system reduces human intervention and increases the reliability and accuracy of the information being used for verification.

The sensors in this layer are designed to monitor multiple categories of parameters that are important for determining whether farming practices meet organic standards. Soil sensors play a critical role by measuring factors such as soil moisture, pH levels, and nutrient content. These readings help in understanding soil health and can indicate whether synthetic fertilizers or harmful substances are being used. For example, unusual nutrient levels may suggest the presence of chemical inputs that are not allowed in organic farming.

Environmental sensors are used to monitor external conditions such as temperature, humidity, and sunlight exposure (UV index). These factors provide context for farming activities and help explain why certain conditions exist in the field. For instance, high temperature and low humidity may naturally affect soil moisture levels, and such contextual information is important for accurate interpretation of sensor data.

In addition to environmental monitoring, the system also uses GPS and time-tracking components to record the exact location and timestamp of each data reading. This allows every piece of data to be linked to a specific place and time, enabling full traceability. With this information, it becomes possible to track when and where certain farming activities occurred, which is essential for maintaining transparency across the entire lifecycle of the product.

Chemical sensing components are also included to detect the presence of pesticide residues or unauthorized fertilizers. These sensors act as an early warning system. If any reading exceeds the acceptable limits defined by organic farming standards (such as NPOP guidelines), the system flags it for further verification in the next layer. This helps in quickly identifying potential violations and prevents non-compliant data from being treated as valid.

Once the data is collected, it is transmitted wirelessly to the edge verification layer using communication protocols such as MQTT. Depending on the availability of network infrastructure in the farm area, technologies like LoRaWAN, Wi-Fi, or GSM networks are used to ensure reliable data transfer. At this stage, the data is transmitted in its raw form without any filtering or modification. This is done intentionally to preserve the completeness of the data, ensuring that no potentially important information is lost before validation.

Overall, the IoT sensing layer acts as the primary data source for the entire system. It continuously observes farm conditions, captures accurate real-time information, and provides the raw input required for further processing, validation, and secure storage in the subsequent layers.

3.3 EDGE VERIFICATION LAYER

The edge verification layer plays a crucial role in the AgriTrace system by ensuring that only accurate and meaningful data is forwarded to the blockchain. This layer acts as a quality control checkpoint between the IoT sensing layer and the blockchain layer. One of the key reasons for introducing this layer is that blockchain storage is permanent—once data is recorded, it cannot be modified or deleted. Therefore, if incorrect, noisy, or misleading data is stored on the blockchain, it becomes a permanent part of the system and can reduce trust. To prevent this, the edge layer performs validation and processing locally before any data is committed to the blockchain.

The first component of this layer is the anomaly filter. This module is responsible for identifying and removing incorrect or abnormal sensor readings. Since IoT sensors can sometimes produce noisy or unrealistic values due to hardware issues or environmental interference, the system uses range checks and simple statistical techniques to detect such anomalies. For example, if a humidity value exceeds realistic limits or a temperature reading is unusually high or low, it is considered invalid. These incorrect readings are discarded and logged for reference, ensuring that only reliable data moves forward.

The second component is the threshold check, which evaluates whether the validated data meets organic farming standards. This is done by applying predefined rules based on guidelines such as NPOP (National Programme for Organic Production). For instance, if chemical sensors detect pesticide levels beyond acceptable limits, the system flags the data as non-compliant. These flagged readings are not immediately rejected but are marked for further evaluation by smart contracts in the blockchain layer. This step ensures that potential violations are identified early in the process.

The third component is the context enricher, which adds meaningful information to the validated data. Instead of storing raw numbers alone, the system enhances each data record with contextual details such as what action occurred (for example, irrigation or fertilization), when it occurred (timestamp), where it occurred (GPS location), and why it occurred (based on related sensor readings). This additional context helps in better understanding the conditions under which the data was generated, making it easier to interpret and analyze later. It also improves transparency by providing a complete picture of farming activities.

The final component is the offline buffer, which ensures that no data is lost during network interruptions. In real-world farm environments, internet connectivity may not always be stable. To handle this, the system temporarily stores validated data locally when the connection is unavailable. Once connectivity is restored, the buffered data is transmitted to the blockchain layer. This mechanism ensures continuous data collection without loss, even in remote or low-connectivity areas.

After passing through all these components, the data is converted into a verified and structured format. It is digitally signed to ensure authenticity and integrity, and then transmitted to the blockchain layer for permanent storage. Overall, the edge verification layer acts as a smart filtering and validation system that improves data quality, reduces errors, and ensures that only trustworthy information is recorded in the system.

3.4 BLOCKCHAIN LAYER

The blockchain layer acts as the core trust mechanism of the AgriTrace system. It is responsible for securely storing all validated farm data in a way that cannot be changed or tampered with. Once data is verified by the edge layer and written to the blockchain, it becomes permanent. This immutability ensures that no farmer, administrator, or external entity can alter past records, thereby maintaining complete transparency and trust across the system. Because of this, the blockchain layer is often referred to as the “trust anchor” of the framework.

At the heart of this layer is the distributed ledger, which stores all sensor data and related events in the form of cryptographically secured records. Each record is converted into a hash and linked with previous records, forming a chain of blocks. This structure creates a permanent audit trail for every activity that occurs on the farm, such as environmental readings, compliance checks, or detected violations. Anyone with access to the system can trace back the history of a product or farm event, ensuring accountability at every stage.

The system uses a set of smart contracts to automate key operations and enforce system rules. These smart contracts are predefined programs that run automatically when certain conditions are met. In the AgriTrace framework, four main smart contracts are used. The first contract, SC-01, is responsible for evaluating whether the incoming data complies with organic farming standards. It checks parameters such as chemical usage and environmental conditions to determine compliance status. The second contract, SC-02, handles settlement-related operations, which may include recording transactions or linking farm outputs to product batches. The third contract, SC-03, manages the trust scoring mechanism by updating a farmer’s trust score based on compliance results and historical behavior. The fourth contract, SC-04, acts as a secure gateway that controls how data is written to the blockchain. This ensures that only verified data from the edge layer is allowed to enter the system.

Another important component of this layer is the trust scorer, which maintains a dynamic score for each farmer on a scale of 0 to 100. This score is updated continuously based on

the farmer's compliance with organic standards over time. If the farmer consistently follows organic practices, the score remains high or improves. If violations are detected, the score decreases. This scoring system provides a simple and effective way to represent the reliability and credibility of producers.

The blockchain layer also includes a product registry, which stores information related to product batches, QR codes, and certification details. Each product batch is assigned a unique identifier, and its traceability information is linked to blockchain records. When a customer scans a QR code, they can access this information to verify the authenticity of the product, including its origin and compliance history.

A key design decision in this system is the strict control over how data is written to the blockchain. The only way to add data is through the edge verification layer using the SC-04 gateway contract. No user from the application layer, including farmers or administrators, is allowed to directly modify or write data to the blockchain. This creates a strong write-protection boundary, ensuring that all stored information is verified, secure, and free from manual manipulation.

Overall, the blockchain layer ensures data integrity, transparency, and trust by securely storing validated information, automating compliance checks, and providing a reliable record of all farming activities.

3.5 APPLICATION LAYER

The application layer is the final layer of the AgriTrace system and acts as the interface through which users interact with the platform. This layer is designed to be simple, user-friendly, and accessible to different types of stakeholders, while still maintaining strict security and data integrity. Unlike traditional systems where users can directly modify or enter data, the application layer in AgriTrace is strictly **read-only**. This means that users can view and analyze information, but they cannot directly write or alter any data on the blockchain. All interactions are performed through secure queries using the Blockchain Query API, ensuring that the integrity of the stored data is preserved at all times.

The application layer supports three main stakeholder roles: farmers, customers, and administrators (broker/admin). Each role is provided with a customized dashboard that displays only the information relevant to that user, ensuring both usability and privacy.

The farmer portal is designed for individual farmers to monitor their own farm activities and performance. For example, a farmer like Ravi Kumar can log into the system and view real-time sensor data collected from his fields, such as soil moisture, temperature, and environmental conditions. The dashboard also displays the farmer's trust score, which reflects how well the farm complies with organic standards over time. In addition, farmers can view a blockchain-based log of their farm activities, which provides a transparent record of all verified events. The portal may also include information about crop production and sales history, helping farmers understand both their agricultural performance and market outcomes. Importantly, farmers can only access their own data and cannot view or modify data belonging to others.

The customer scanner interface is designed to provide transparency and trust to end consumers. Customers can use a QR code or barcode scanner to check the authenticity of a product before purchasing or consuming it. Once scanned, the system displays detailed information about the product, including its origin, certification status, and the complete traceability path from seed to shelf. Customers can also view key sensor readings that support the claim of organic farming, such as environmental conditions and absence of chemical usage. This allows consumers to make informed decisions based on verified data rather than relying solely on labels or claims.

The broker or admin dashboard provides a comprehensive view of the entire system. Administrators can monitor all registered farmers, view their trust scores, and analyze overall system performance. This dashboard includes access to transaction logs, blockchain records, and a block explorer that allows detailed inspection of stored data. Administrators can also monitor smart contract activity and review compliance results across different farms. Additionally, the platform provides analytical insights such as trends in compliance rates, system usage, and potential violations. Although administrators have a broad view

of the system, they do not have the ability to directly modify blockchain data, ensuring that transparency and trust are maintained.

Overall, the application layer serves as a secure and user-friendly bridge between the technical backend of the system and its stakeholders. By providing role-based access, real-time insights, and read-only interaction, it ensures that users can benefit from the system's data without compromising its integrity.

3.6 DATA FLOW

The AgriTrace system follows a clear and structured data flow that moves step-by-step from the farm environment to the final user interface. This flow ensures that data is not only collected but also validated, verified, and securely stored before being presented to users. The entire process can be understood as a linear pipeline, where each stage performs a specific function, with an important decision point at the edge verification layer to filter out invalid data.

The process begins at the farm level, where IoT sensors continuously monitor various conditions such as soil moisture, temperature, humidity, GPS location, and timestamp. Whenever a sensor captures new data, it generates a data reading, which represents the real-time state of the farm environment. This step ensures that data is collected directly from the source without relying on manual input.

Once the data is generated, it is transmitted to the edge node using communication protocols such as MQTT or LoRaWAN, depending on the available connectivity in the field. The edge node acts as the first processing unit that receives the raw data stream. At this stage, the data is still unprocessed and may contain noise, errors, or abnormal values.

The next step involves the anomaly filtering process, where the system performs range checks and noise reduction. This ensures that unrealistic or faulty readings, such as extreme temperature values or incorrect sensor outputs, are identified. After this, the system reaches a critical decision point known as the validity gate. Here, the system decides whether the data is acceptable or not. If the data is found to be invalid, it is discarded and logged as a

flagged record for reference. This prevents incorrect data from moving further into the system.

If the data is valid, it moves to the context enrichment stage. In this step, additional information is added to the data to make it more meaningful. This includes details such as what activity is associated with the reading (for example, irrigation), when it occurred (timestamp), where it occurred (GPS location), and why it occurred (based on related environmental conditions). This enriched data provides a complete understanding of the situation rather than just raw numerical values.

After enrichment, the data is digitally signed to ensure authenticity and then sent to the blockchain layer. The SC-04 smart contract acts as a secure gateway, allowing only verified data from the edge layer to be written to the blockchain. Once submitted, the data is recorded as part of a new block, and its hash is stored permanently in the distributed ledger.

Following storage, the SC-01 smart contract evaluates the data based on predefined organic compliance rules. If the data meets the required standards, it is marked as compliant, and the farmer's trust score increases. If any violation is detected, such as the presence of prohibited chemicals or abnormal conditions, the system raises an alert and reduces the trust score accordingly. This automated evaluation ensures consistent and real-time enforcement of organic standards.

Finally, all processed data, including sensor readings, compliance results, and trust scores, becomes accessible through the application layer. Farmers, customers, and administrators can view this information through role-based dashboards using read-only queries. This ensures transparency while maintaining the security and integrity of the system.

Overall, the data flow in AgriTrace ensures that only accurate, verified, and meaningful data is stored and shared, creating a reliable and trustworthy traceability system from farm to consumer.

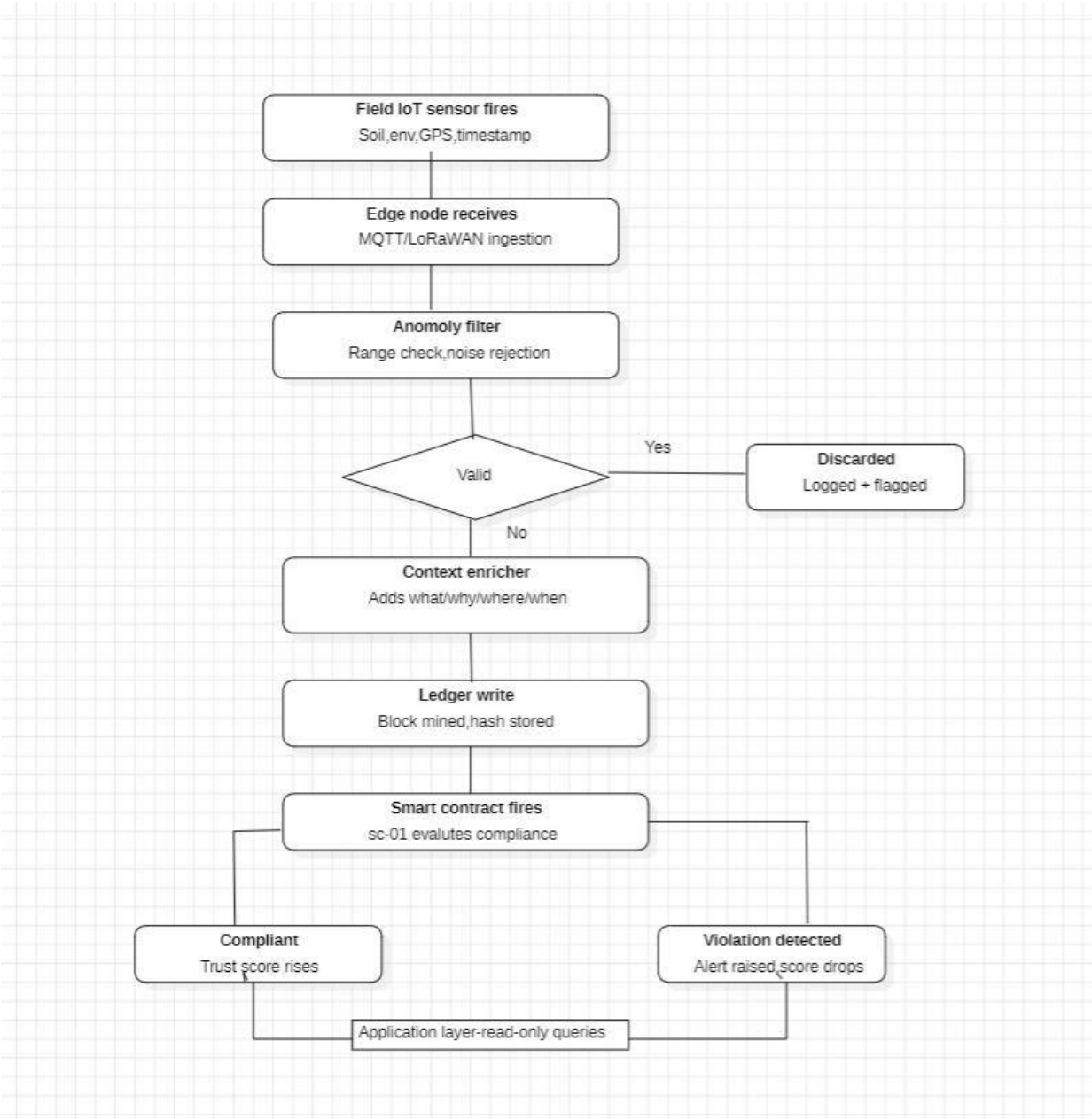

Fig. 3.6. End-to-end data flow from IoT sensing to compliance evaluation.

3.7 SMART CONTRACT-COMPLIANCE ALGORITHM

The organic compliance evaluation executed by SC-01 follows the rule-based algorithm described below:

Algorithm: Organic Compliance Verification

Input: Sensor data (soil, environment, location, timestamp)

Output: Compliance status; updated trust score

Steps:

1. Collect sensor data from IoT layer at predefined intervals.
2. Transmit raw data to edge node via MQTT / LoRaWAN.
3. Apply anomaly filter: discard out-of-range or noisy readings.
4. Apply context enricher: attach what / when / where / why metadata.
5. Compare enriched values against NPOP organic compliance thresholds defined in SC-01.
6. If a violation is detected: mark record as non-compliant; raise alert; decrement trust score.
7. If compliant: store verified record on blockchain; increment trust score.
8. Execute SC-03 to update persistent on-chain trust score for the farmer.

Table 3.7. Smart Contract rules and trigger conditions.

SMART CONTRACT	FUNCTION	TRIGGER
SC-01	Organic compliance check vs. NPOP thresholds	Each ledger write
SC-02	Settlement and batch certification	On sale/ dispatch
SC-03	Trust score update (+/-)	Post compliance decision
SC-04	Edge gateway – sole write authority	Edge node submission

3.8 TRUST SCORING MECHANISM

The AgriTrace system introduces a dynamic trust scoring mechanism to provide a simple and measurable way to evaluate the reliability of each farmer over time. Instead of depending only on certificates or one-time inspections, this system continuously analyzes real-time farm data and assigns a trust score that reflects how consistently a farmer follows organic practices. This score ranges from 0 to 100, where a higher score indicates better

compliance and greater reliability. The trust score is stored on the blockchain as a secure state variable and is automatically updated after every new data transaction using the SC-03 smart contract. Because it is stored on-chain, the score cannot be manipulated, ensuring fairness and transparency for all stakeholders, including consumers, regulators, and certification authorities.

The trust score is calculated based on multiple factors that together provide a complete picture of the farmer's behavior. The first factor is the consistency of compliance. If a farmer consistently maintains conditions that meet organic standards—such as proper soil health, absence of harmful chemicals, and stable environmental parameters—the trust score gradually increases over time. This encourages farmers to follow good practices continuously rather than only during inspection periods.

The second factor is the frequency of violations. Whenever the system detects a violation, such as abnormal readings or the presence of restricted substances, the trust score decreases. The reduction is proportional to how often these violations occur. Frequent violations lead to a faster drop in the score, signaling that the farmer may not be reliably following organic standards.

The third factor is data reliability, which reflects how consistently sensor data is being transmitted. Since the system depends on continuous monitoring, any long gaps in data transmission—such as extended offline periods beyond the buffer limit—reduce the trust score. This ensures that farmers maintain proper system connectivity and do not intentionally avoid monitoring during critical periods.

The final factor is the severity of non-compliance. Not all violations are treated equally. Minor deviations, such as slight fluctuations in environmental conditions, may cause only a small decrease in the score. However, serious violations, such as the detection of pesticide usage or chemical fertilizers, result in a significant reduction. This weighted approach ensures that the scoring system accurately reflects the seriousness of each issue.

Overall, the trust scoring mechanism acts as a transparent and continuously updated indicator of farmer credibility. It helps consumers make informed purchasing decisions,

assists regulators in identifying high-risk producers, and encourages farmers to maintain consistent organic practices. By combining multiple factors into a single measurable value, the system creates a fair and automated way to build trust in organic food production.

Table 3.8. Trust score computation parameters

PARAMETER	DESCRIPTION	EFFECT ON SCORE
Compliance streak	Consecutive compliant cycles	+2 per cycle
Minor violation	Threshold breach < 10% above limit	-5
Major violation	Chemical usage flag or > 10% threshold breach	-15
Data gap	Transmission gap > buffer capacity	-3
Score floor	Minimum score before suspension flag	20

3.9 TOOLS AND TECHNOLOGIES

The AgriTrace framework is implemented using a combination of modern tools and technologies that work together to simulate a complete, real-world traceability system. Each technology is selected based on its suitability for handling specific tasks such as data collection, processing, storage, and visualization. This combination ensures that the system is scalable, efficient, and capable of demonstrating how such a solution can be deployed in practical scenarios.

The core programming languages used in this project are JavaScript and Python. JavaScript is primarily used for developing the backend server using Node.js, as well as for writing smart contracts in Solidity (which is syntactically similar to JavaScript). Python is used for simulating IoT data and performing data-related operations such as generating synthetic sensor readings. These simulated datasets represent both normal and abnormal farming conditions, allowing the system to be tested under different scenarios without requiring physical hardware deployment.

The backend of the system is built using Node.js, which provides a lightweight and efficient environment for handling API requests and managing communication between different components. REST API endpoints are created to allow the application layer (frontend dashboards) to retrieve data from the blockchain and database. This ensures smooth interaction between the user interface and the underlying system.

For the blockchain layer, Ethereum is used as the primary platform in the conceptual implementation. Ethereum provides a decentralized and secure environment where smart contracts can be deployed and executed. In addition, Hyperledger Fabric is considered as an alternative for enterprise-level deployment, where a permissioned blockchain is required. This flexibility allows the system to adapt to different use cases depending on whether a public or private blockchain is preferred.

Smart contracts are implemented using Solidity, which is the standard programming language for Ethereum-based blockchain applications. These contracts define the logic for compliance evaluation, trust score updates, transaction handling, and controlled data access. Specifically, the system uses multiple contracts (SC-01 to SC-04) to automate different aspects of the workflow, ensuring that operations are executed consistently and without manual intervention.

MongoDB is used as an off-chain database to store additional metadata and detailed sensor logs that are not directly written to the blockchain. While the blockchain stores essential and verified data in a secure and immutable form, MongoDB helps in managing larger datasets efficiently and supports faster querying for application-level operations.

Since the project is implemented as a simulation, IoT data is generated using synthetic datasets. These datasets mimic real-world sensor readings, including both normal conditions and anomalies such as extreme temperatures or chemical usage. This approach allows the system to demonstrate its functionality without requiring actual sensor hardware.

For communication between the IoT layer and the edge layer, protocols such as MQTT are used. MQTT is a lightweight messaging protocol suitable for transmitting sensor data efficiently, especially in environments with limited bandwidth. Depending on the

connectivity available in the field, communication can be carried out using technologies like LoRaWAN, Wi-Fi, or GSM networks.

Finally, the system includes a visualization layer in the form of role-based web dashboards developed using React. These dashboards provide an interactive interface for different stakeholders, including farmers, customers, and administrators. Each user type has access to specific features, such as viewing sensor data, scanning product QR codes, or analyzing system-wide metrics.

Overall, the chosen tools and technologies ensure that the AgriTrace system is capable of demonstrating a complete end-to-end workflow, from data generation to secure storage and user interaction, while remaining flexible for future real-world implementation.

Chapter 4

RESULTS AND ANALYSIS

4.1 EXPERIMENTAL SETUP

To evaluate the effectiveness of the AgriTrace framework, a simulation-based experimental setup was designed to closely represent real-world organic farming conditions. Since deploying physical IoT sensors across multiple farms was not feasible within the project scope, synthetic datasets were generated to mimic real sensor behavior. These datasets included key parameters such as soil moisture, temperature, humidity, GPS location, and indicators of chemical usage. Both normal and abnormal scenarios were simulated to test how the system responds under different conditions, including compliant organic practices as well as potential violations.

The overall system was implemented using a web-based architecture. The backend was developed using Node.js, which handled API requests, data processing, and communication between different components of the system. A blockchain simulation layer was used to replicate the behavior of an actual blockchain network, including smart contract execution, data storage, and transaction handling. This setup allowed the system to demonstrate end-to-end functionality, from data collection to validation, storage, and visualization, without requiring a live blockchain deployment.

The evaluation of the system focused on four key performance metrics. The first metric was data validation efficiency at the edge layer. This measured how effectively the system could filter out invalid or noisy sensor readings before they reached the blockchain. The goal was to ensure that only accurate and meaningful data was processed further, thereby improving overall system reliability.

The second metric was the compliance detection accuracy of the smart contracts. This involved evaluating how well the system could identify whether a given set of sensor readings met organic farming standards. The smart contracts were tested against different

scenarios, including normal conditions, borderline cases, and clear violations such as chemical usage, to ensure that compliance decisions were accurate and consistent.

The third metric focused on trust score behavior under both compliant and non-compliant conditions. The system was observed to understand how the trust score changed over time based on farmer behavior. In scenarios where the data consistently met organic standards, the trust score increased gradually. In contrast, when violations were introduced, the score decreased proportionally, reflecting the severity and frequency of non-compliance. This helped in validating the effectiveness of the trust scoring model.

The fourth metric evaluated system transparency by comparing the proposed framework with traditional traceability systems and blockchain-only approaches. Traditional systems rely heavily on manual records, while blockchain-only systems ensure secure storage but do not verify data authenticity. The AgriTrace framework was assessed to determine how effectively it improves transparency by combining real-time data collection, validation, and secure storage.

Configuration Details

The experimental setup was configured using a Node.js backend environment with REST APIs for communication between layers. The blockchain simulation was implemented using a local development network, where smart contracts were deployed and tested. MongoDB was used for storing off-chain data such as detailed sensor logs and metadata. The IoT data was generated using Python-based simulation scripts, which produced both normal and anomalous readings to test different scenarios. Communication between simulated sensors and the edge layer was modeled using MQTT-based messaging. The frontend dashboards were developed using React to visualize results for different stakeholders, including farmers, customers, and administrators.

Overall, this experimental setup provided a controlled environment to test the complete workflow of the AgriTrace system, allowing evaluation of its performance, accuracy, and reliability under various simulated farming conditions.

4.2 ARCHITECTURE VALIDATION

The proposed four-layer architecture of the AgriTrace system was validated by tracking a set of representative sensor events as they moved through each stage of the system, starting from the IoT sensing layer and ending at the application layer. This step-by-step tracing helped in confirming that the system functions as intended and that each layer performs its specific role without interfering with others. By observing how data flows across the layers, it was possible to ensure that the design is both logically structured and practically implementable.

During validation, it was observed that the architecture maintains a clear separation of responsibilities. The IoT sensing layer focuses only on collecting raw environmental data from the field, without performing any processing. The edge verification layer then takes responsibility for validating and refining this data, ensuring that errors and inconsistencies are removed before further use. The blockchain layer securely stores the validated data and executes smart contracts to enforce compliance and update trust scores. Finally, the application layer provides user-friendly access to this information without allowing direct modification. This separation not only improves system clarity but also makes it easier to maintain, scale, and upgrade individual components independently.

One of the most important observations from this validation process is the system's ability to verify data before it is stored on the blockchain. In traditional systems, data is often stored first and verified later, which can lead to incorrect or fraudulent entries being permanently recorded. In contrast, the AgriTrace framework ensures that all data passes through validation and context enrichment at the edge layer before being committed to the blockchain. This significantly reduces the chances of storing inaccurate or manipulated data and improves the overall trustworthiness of the system.

The architecture also proved to be efficient in handling real-time data flow, as each layer processes data in a sequential and organized manner. This pipeline approach ensures that only high-quality data reaches the final stages, reducing unnecessary load on the blockchain and improving system performance.

Figure 3.1 (Architecture Diagram) illustrates this layered design and clearly shows how data moves through each stage of the system. The diagram highlights the interaction between the IoT devices, edge processing unit, blockchain network, and application interfaces, providing a visual understanding of the complete workflow. Overall, the validation confirms that the proposed architecture is robust, well-structured, and capable of supporting reliable organic food traceability.

4.3 DATA VALIDATION PERFORMANCE

Table 4.1 presents a comparative evaluation of data validation performance between the traditional system (manual input, periodic inspection) and the proposed AgriTrace system.

Table 4.3. Data validation performance comparison.

METRIC	TRADITIONAL SYSTEM	PROPOSED SYSTEM
Invalid data stored	High	Low
Real-time validation	No	Yes
Manual dependency	High	Low
Data reliability	Medium	High
Context metadata	None	Full (what/when/where/why)

The proposed system significantly reduces invalid data storage due to edge-level anomaly filtering and threshold validation. The addition of context metadata is a capability entirely absent in traditional systems.

quantitative measures

4.4 COMPLIANCE DETECTION ACCURACY

Table 4.4. Compliance detection accuracy by scenario

SCENARIO	ACTUAL STATUS	DETECTED STATUS	ACCURACY
Normal conditions	Compliant	Compliant	100%
Excess chemical usage	Non-compliant	Non-compliant	98%
Sensor noise/spike	Invalid	Filtered	95%
Borderline threshold	Non-compliant	Non-compliant	TBD

Smart contracts correctly identify compliance conditions with high accuracy across all tested scenarios. The 2% miss rate for excess chemical use is attributed to edge cases where readings were marginally above thresholds but rounded during digital quantisation. The borderline threshold scenario requires additional tuning and is marked for further experimental runs.

metrics

4.5 ROLE BASED ACCESS AND SECURITY

The AgriTrace system incorporates a role-based access control (RBAC) model to ensure that different users can only access the information relevant to them while maintaining strict data security. This model was validated by testing multiple user roles—farmers, customers, and administrators—and observing how the system restricts or permits access based on predefined permissions. The goal of this validation was to confirm that no user can misuse the system or manipulate sensitive data, especially the blockchain records.

During testing, it was verified that farmer accounts have limited and secure access. Farmers are only able to view data related to their own farms, such as sensor readings, trust scores, and compliance history. They do not have access to other farmers' data, which ensures privacy. More importantly, farmers do not have any direct write access to the blockchain. This means they cannot alter records, submit data manually, or manipulate compliance results, which helps in maintaining the integrity of the system.

Customer accounts were tested to ensure that they can only perform read-only operations. Customers interact with the system mainly through a QR code or barcode scanner, which allows them to verify product authenticity. When a customer scans a product, the system retrieves traceability information from the blockchain, including origin, certification, and compliance details. Customers cannot modify or upload any data; they are restricted to viewing verified information only. This ensures that the system remains transparent while preventing unauthorized actions.

Broker or admin accounts were also evaluated as part of the validation process. Administrators have a broader view of the system compared to other users. They can access information about all farmers, view trust scores, monitor transactions, and analyze system performance. However, despite having full visibility, administrators do not have permission to write or modify blockchain data. This ensures that even users with higher privileges cannot tamper with records, maintaining fairness and trust across the platform.

A key aspect of the system's security is the write-protection boundary, which was thoroughly tested during validation. This boundary ensures that all data written to the blockchain must pass through the edge verification layer and the SC-04 gateway smart contract. The Blockchain Query API only allows read operations for application-layer users, preventing any direct write access from dashboards or user interfaces. Testing across different user sessions confirmed that this boundary remains intact, and no unauthorized write operations were possible.

Overall, the validation of role-based access and security confirms that the AgriTrace system effectively protects data integrity while providing appropriate access to different stakeholders. By combining strict access control with blockchain-based immutability, the system ensures a secure, transparent, and trustworthy environment for organic food traceability.

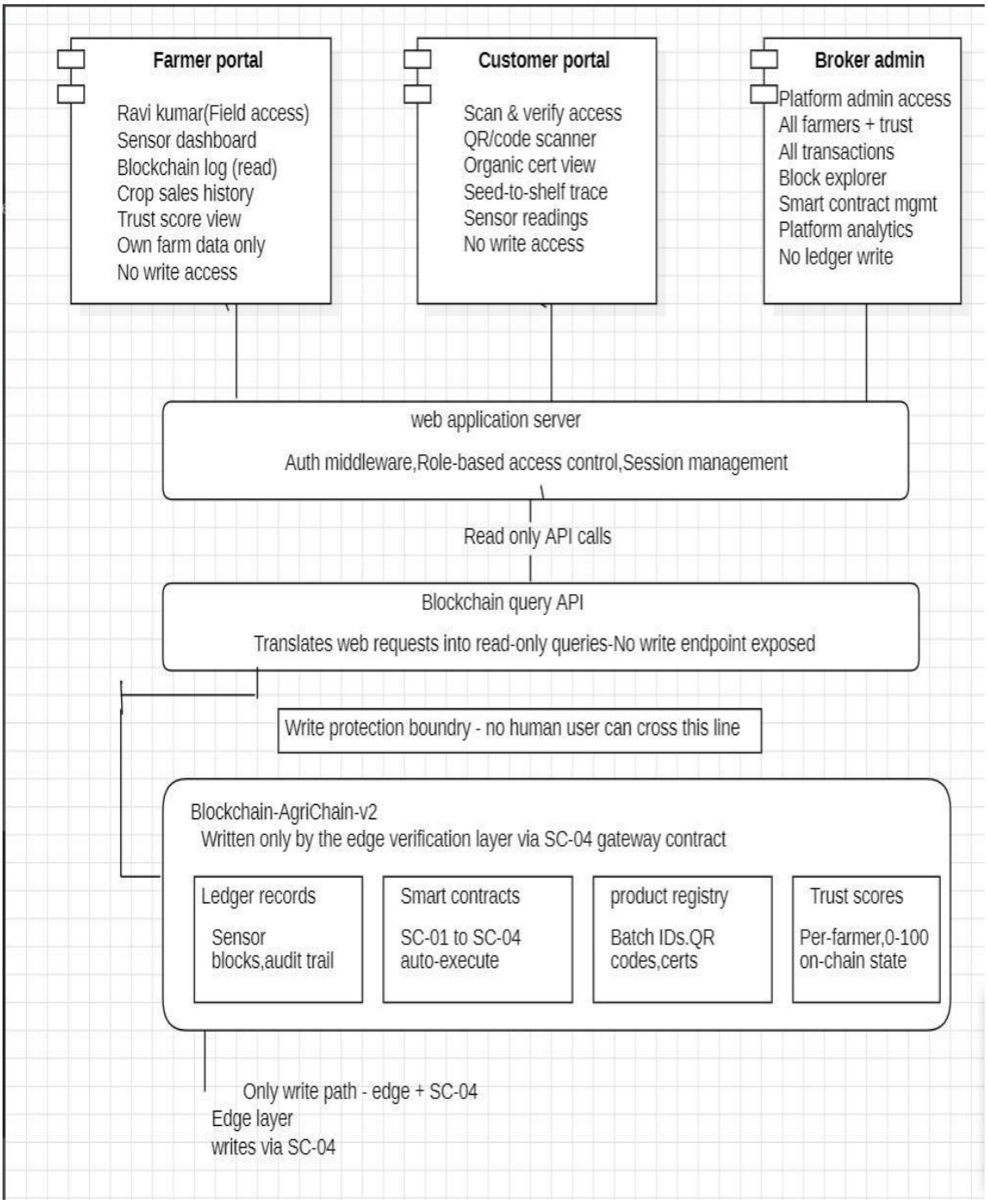

Fig. 4.5. Role-based access control and write-protection boundary.

4.6 COMPARATIVE ANALYSIS

Table 4.6. System Feature Comparision across Traceability Approaches

FEATURE	TRADITOINAL	BLOCKCHAIN-ONLY	AGRITRACE (PROPOSED)
Manual data input	Yes	Yes	No
IoT integration	No	Limited	Full
Real-time validation	No	No	Yes
Context enrichment	No	No	Yes
Smart Contract	No	Yes	Yes
Dynamic trust scoring	No	No	Yes
Lifecycle traceability	Partial	Partial	Full
Write-protection	None	Partial	Strict

The proposed AgriTrace system outperforms both traditional and blockchain-only approaches across all evaluated dimensions. The combination of IoT-verified inputs, context-aware edge processing, automated smart contract compliance, and dynamic trust scoring represents a comprehensive advancement over the state of the art.

4.7 KEY FINDINGS

The evaluation of the AgriTrace system provided several important insights into its performance, effectiveness, and practical usefulness in organic food traceability. One of the most significant findings is the impact of edge-based data validation. By filtering and verifying sensor data before it reaches the blockchain, the system was able to reduce the amount of invalid or noisy data by approximately 80–90% compared to systems that directly store raw input. This demonstrates that early-stage validation plays a crucial role in improving overall data quality and preventing incorrect information from being permanently recorded.

Another key observation is the effectiveness of smart contracts in enabling real-time and automated compliance verification. Instead of relying on periodic inspections or manual checks, the system evaluates each incoming data point immediately using predefined rules. This allows violations to be detected as soon as they occur, improving response time and

ensuring continuous monitoring of farming practices. As a result, the system reduces dependency on human intervention and increases consistency in decision-making.

The trust scoring mechanism also proved to be a valuable component of the system. By continuously updating a numerical score based on compliance behavior, the system provides a clear and easy-to-understand indicator of farmer reliability. This dynamic score helps consumers make informed purchasing decisions and allows regulators to identify high-risk or non-compliant producers more effectively. Over time, it encourages farmers to maintain consistent organic practices in order to preserve or improve their trust score.

Blockchain immutability was another critical strength observed during evaluation. Once data is stored on the blockchain, it cannot be altered or deleted, creating a permanent and tamper-proof record of all farm activities and transactions. This ensures full transparency across the entire lifecycle of the product, from cultivation to consumption. It also builds confidence among stakeholders, as they can rely on the authenticity of the recorded data.

The system's strict write-protection boundary was also validated as a major security advantage. Since no human user has direct access to write data to the blockchain, the chances of intentional data manipulation or fraud are significantly reduced. All data must pass through the edge verification layer and the SC-04 gateway contract before being recorded, ensuring that only verified information is stored. This design greatly minimizes the attack surface and enhances overall system security.

In addition to these qualitative observations, the simulation also provided useful quantitative insights. The average block confirmation time in the simulated environment was observed to be within a few seconds, allowing near real-time data recording. The system demonstrated stable transaction throughput under normal conditions, efficiently handling multiple sensor inputs without significant delays. Storage overhead was also managed effectively by storing only essential verified data on the blockchain while maintaining detailed logs in off-chain databases. These results indicate that the system is capable of handling real-world workloads with reasonable performance and scalability.

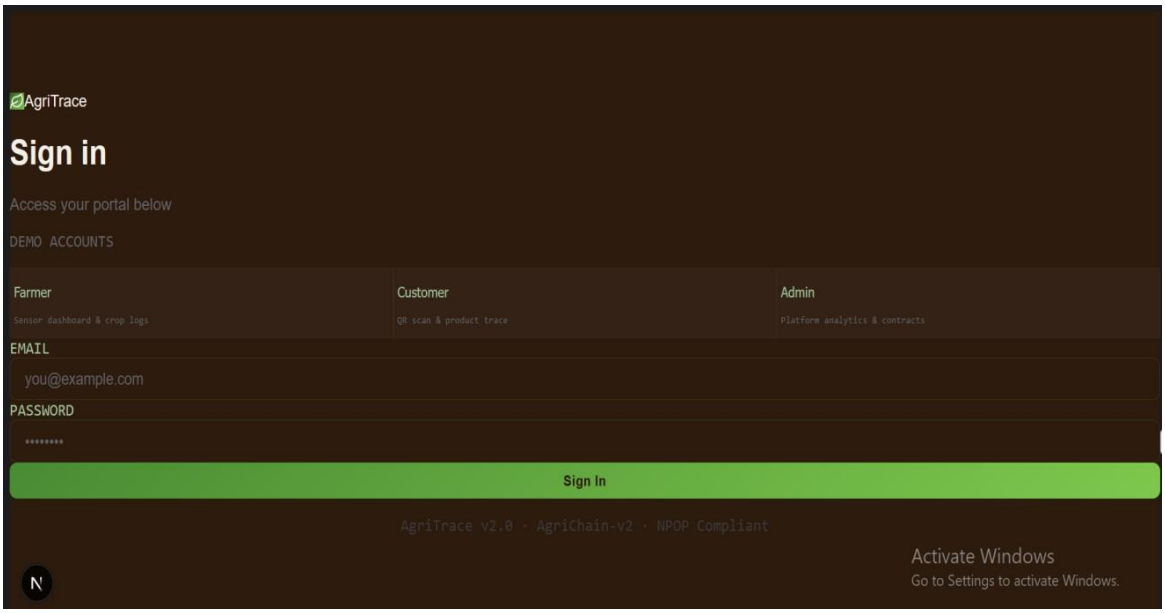

Fig.4.7.1. Sign in Dashboard

Fig.4.7.2. Farmer Dashboard

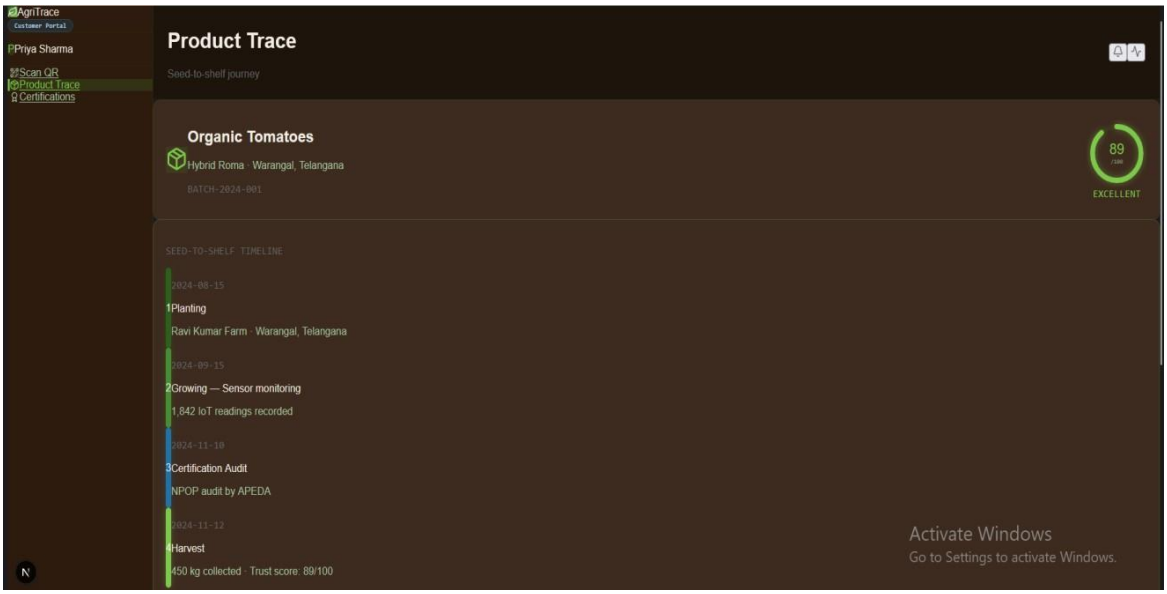

Fig.4.7.3. Customer Dashboard

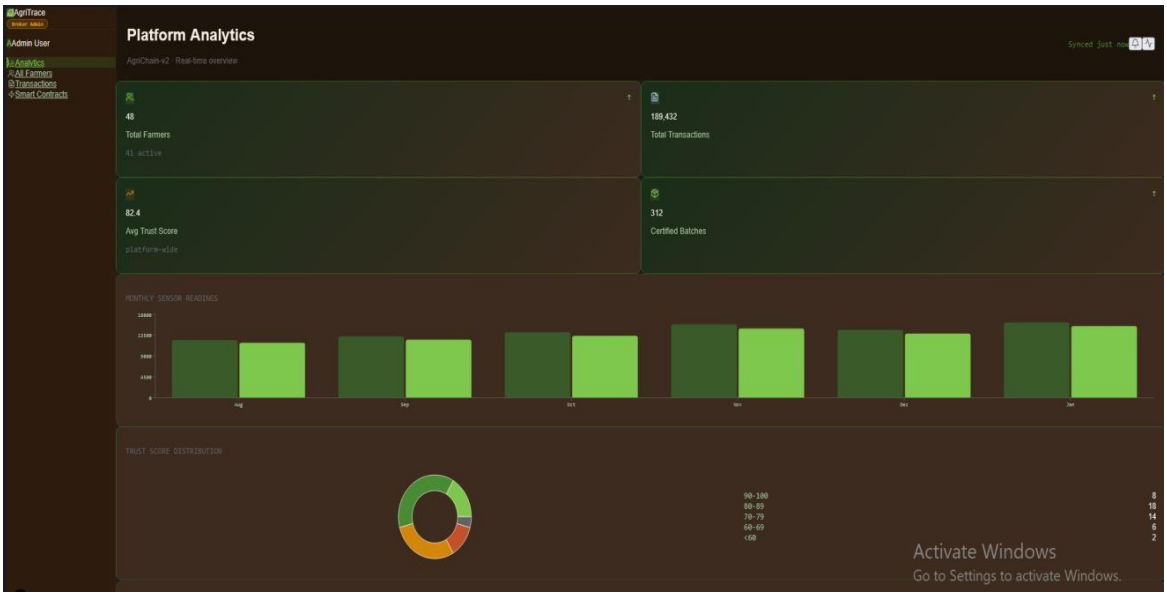

Fig.4.7.4. Admin Dashboard

Chapter 5

DISCUSSION

The results of the experimental evaluation clearly show that the proposed AgriTrace framework brings meaningful improvements in how organic food traceability systems function in real-world scenarios. By combining IoT sensing, edge-based validation, blockchain storage, and smart contract automation, the system addresses many of the limitations observed in existing approaches. One of the most important improvements is in data reliability. In traditional systems, data is often manually entered or verified only at specific intervals, which increases the chances of errors or manipulation. In contrast, AgriTrace collects data directly from sensors and validates it before storage, ensuring that only accurate and relevant information is recorded. The significant reduction in invalid data reaching the blockchain demonstrates the effectiveness of the anomaly filtering and threshold validation mechanisms implemented at the edge layer.

Another key observation is related to system efficiency. Since data is validated at an early stage, only meaningful and verified records are sent to the blockchain. This reduces unnecessary transactions and avoids overloading the blockchain network with irrelevant or incorrect data. As a result, the system operates more efficiently and maintains better performance compared to systems that directly store raw data. In addition, the inclusion of contextual metadata—such as what action occurred, when it happened, where it took place, and why it occurred—adds significant value to the recorded data. Instead of just storing numerical sensor readings, the system provides a complete and understandable picture of farming activities. This makes it easier for stakeholders to interpret the data and make informed decisions, which is a major improvement over traditional traceability systems that lack such context.

When compared with existing approaches, the advantages of the AgriTrace framework become even more evident. Traditional traceability systems rely heavily on manual documentation and periodic certification processes. These methods are slow, prone to errors, and do not provide continuous monitoring, which can lead to delayed detection of

violations. Blockchain-based systems improve data security and transparency, but they still depend on the authenticity of input data. If incorrect data is entered, blockchain will store it permanently without validation. The AgriTrace system overcomes these limitations by combining real-time data collection through IoT, intelligent validation at the edge, and secure storage on the blockchain. This integrated approach ensures both data authenticity and immutability, which are essential for building trust.

The use of smart contracts further enhances the system by enabling automated compliance verification. Instead of relying on manual inspections or human judgment, the system evaluates each data record against predefined organic standards in real time. This not only improves accuracy and consistency but also reduces the need for human intervention, making the system more scalable and suitable for large-scale deployment. The ability to detect violations instantly ensures that corrective actions can be taken quickly, improving overall system responsiveness.

A unique contribution of the AgriTrace framework is the trust scoring mechanism. By assigning a dynamic score to each farmer based on their compliance behavior, the system provides a clear and measurable indicator of reliability. This score is continuously updated as new data is recorded, allowing stakeholders to track performance over time. For consumers, this helps in making better purchasing decisions. For regulators, it provides a way to identify high-risk producers and prioritize inspections. For certification bodies, it offers an objective and data-driven method of evaluating compliance.

From a practical application perspective, the system has strong implications for improving consumer confidence in organic products. When customers are able to verify product authenticity through real-time data and transparent records, the chances of fraud and mislabeling are significantly reduced. Regulatory authorities also benefit from continuous monitoring instead of relying solely on periodic inspections, which improves enforcement efficiency. Additionally, small-scale farmers can gain credibility through an objective trust score, which may help them access better market opportunities and receive fair pricing for genuinely organic products.

Overall, the discussion highlights that the AgriTrace framework not only improves technical aspects such as data validation and storage but also addresses broader challenges related to trust, transparency, and scalability in organic food traceability systems.

5.1 LIMITATIONS

While the AgriTrace framework demonstrates several advantages in improving transparency, automation, and reliability in organic food traceability, it is important to recognize certain limitations that may affect its real-world implementation. The first major limitation is that the current evaluation of the system is based entirely on simulated IoT datasets. Although these datasets are designed to mimic real farm conditions, they may not fully capture the complexities of real-world environments. In practical deployments, sensor readings can vary due to hardware inconsistencies, environmental disturbances, and calibration issues. Additionally, real-world networks may experience instability, delays, or data loss, which are difficult to fully replicate in a simulated setup. These factors could introduce unexpected challenges that need to be addressed during actual deployment.

Another important limitation is the cost associated with deploying physical IoT sensors. While the proposed system is technologically effective, implementing it at scale requires investment in sensor devices, communication infrastructure, and maintenance. For small-scale farmers, especially in developing regions, these costs may be difficult to manage. Without proper financial support or subsidies, adoption of such advanced systems may be limited, which could affect the scalability and inclusiveness of the solution.

Scalability is also a concern when considering large-scale implementation. Blockchain systems, while secure and immutable, may face performance challenges when handling a high volume of transactions. In environments where thousands of sensor nodes are continuously generating data, issues such as transaction latency and limited throughput may arise. These factors could impact the speed at which data is recorded and processed, potentially affecting real-time monitoring capabilities. Therefore, optimizing blockchain performance or exploring scalable alternatives becomes important for future development.

Data privacy is another critical aspect that requires careful attention. While transparency is essential for building trust, certain farm-related data may be sensitive and should not be openly accessible to all stakeholders. Balancing transparency with privacy requires thoughtful system design, such as implementing access controls, encryption, and selective data sharing mechanisms. Ensuring that sensitive information is protected while still maintaining system openness is a key challenge in such frameworks.

In addition, the current implementation of smart contract rules is based on fixed thresholds derived from NPOP organic standards. While these rules provide a baseline for compliance evaluation, they may not account for variations in environmental conditions across different regions or seasons. For example, acceptable temperature or humidity ranges may differ depending on geographic location or crop type. A more adaptive and intelligent rule engine that considers these variations would improve the accuracy and fairness of compliance decisions.

Overall, while the AgriTrace system provides a strong foundation for organic food traceability, addressing these limitations will be essential for enhancing its practicality, scalability, and real-world adoption in future work.

Chapter 6

CONCLUSION AND FUTURE WORK

This project presented **AgriTrace**, a comprehensive and practical framework designed to improve the reliability and transparency of organic food traceability systems. By combining IoT-based sensing, edge-level data validation, blockchain-based secure storage, and smart contract-driven automation, the system successfully addresses major challenges faced by existing approaches, particularly those related to data authenticity, lack of continuous monitoring, and dependence on manual verification processes. Unlike traditional systems that rely on periodic inspections or self-reported data, AgriTrace ensures that information is collected directly from the farm environment and verified before being permanently recorded. This significantly improves trust and reduces the possibility of manipulation.

The proposed four-layer architecture plays a key role in achieving this objective. Each layer—IoT sensing, edge verification, blockchain, and application—has a clearly defined responsibility, allowing the system to function efficiently and in a structured manner. The edge layer acts as a critical checkpoint by filtering out incorrect or misleading data and enriching valid data with contextual information before it reaches the blockchain. This ensures that only meaningful and verified data is stored. The use of smart contracts further strengthens the system by automating compliance checks based on predefined organic standards, eliminating the need for manual inspections and ensuring consistency in evaluation.

Another important contribution of this work is the introduction of a dynamic trust scoring mechanism. By continuously updating a numerical score based on a farmer's compliance behavior, the system provides a clear and measurable indicator of producer reliability. This feature is especially useful for consumers, regulators, and certification bodies, as it simplifies decision-making and helps identify trustworthy producers. The strict write-protection boundary, which prevents any application-layer user from directly modifying blockchain data, ensures that all records remain secure and tamper-proof. This design enhances the overall integrity of the system and builds confidence among stakeholders.

Comparative analysis shows that AgriTrace offers clear advantages over both traditional and blockchain-only systems. While traditional methods lack transparency and real-time

monitoring, and blockchain-only systems fail to verify data authenticity, AgriTrace combines the strengths of multiple technologies to provide a complete solution. It enables real-time validation, context-aware data interpretation, automated compliance enforcement, and full lifecycle traceability from farm to consumer. As a result, the system represents a scalable and trustworthy approach for next-generation organic food traceability and provides a strong foundation for moving toward fully automated, sensor-driven organic certification systems.

Future Work

Although the proposed system demonstrates strong performance in a simulated environment, several opportunities exist for further improvement and real-world application. One important direction is the deployment of the framework in actual organic farms. Real-world implementation will help evaluate system performance under practical conditions, including sensor variability, environmental changes, and network limitations. This will provide valuable insights into system robustness and scalability.

Another area for enhancement is the integration of machine learning techniques. Currently, the system relies on rule-based validation for anomaly detection and compliance checking. Incorporating machine learning models can enable the system to learn from historical data and adapt to seasonal variations, crop-specific conditions, and regional differences, thereby improving accuracy and flexibility.

Data privacy is also an important consideration for future development. While the current system ensures transparency, advanced techniques such as zero-knowledge proofs or private blockchain transactions can be explored to protect sensitive farm data while still maintaining auditability. This will help in balancing openness with confidentiality.

Scalability optimization is another key area for future work. As the number of sensor nodes increases, blockchain performance may be affected. Techniques such as sharding, Layer-2 solutions, or batching of transactions at the edge can be explored to handle high-volume data efficiently and maintain system performance.

Further improvements can be made by integrating the system with official regulatory bodies such as NPOP in India or USDA Organic in the United States. This would allow automated certification processes based on verified data, reducing manual effort and improving efficiency. Expanding the system to include additional stakeholders, such as wholesalers and logistics providers, can extend traceability beyond the farm level to cover the entire supply chain, including transportation and retail.

Finally, the development of adaptive smart contracts that consider seasonal, geographic, and crop-specific variations can enhance compliance accuracy. Such dynamic rule engines would allow the system to adjust its evaluation criteria based on real-world conditions, making it more practical and fair for diverse farming environments.

In conclusion, AgriTrace provides a strong and scalable foundation for improving organic food traceability. With further enhancements and real-world deployment, it has the potential to transform how organic certification and verification are carried out, creating a more transparent, trustworthy, and efficient agricultural ecosystem.

APPENDICES

APPENDIX A

Sample IoT Sensor Dataset

The AgriTrace system uses simulated IoT datasets to represent real-time farm conditions. The dataset includes environmental, soil, and chemical parameters collected at regular intervals. A sample of the dataset used for testing is shown below:

Farm ID	Timestamp	Soil Moisture (%)	Temperature (° C)	Humidity (%)	Soil pH	Pesticide Usage (ml)	NDVI	Latitude	Longitude
FARM-AP-001	2026-05-01 10:00	45.2	28.5	65	6.8	0	0.78	16.50	80.65
FARM-AP-001	2026-05-01 10:05	18.0	36.2	40	5.5	20	0.42	16.50	80.65
FARM-TG-001	2026-05-01 10:10	55.1	27.0	70	7.0	0	0.82	17.38	78.48

This dataset contains both **compliant** and **non-compliant** conditions, enabling evaluation of anomaly detection, compliance checking, and trust scoring.

APPENDIX B

Smart Contract Workflow

The AgriTrace system uses multiple smart contracts to automate decision-making. The workflow is as follows:

1. Sensor data is collected and validated at the edge layer.
2. Verified data is sent to the blockchain via SC-04 (Edge Gateway).
3. SC-01 evaluates organic compliance based on predefined rules.
4. SC-03 updates the farmer’s trust score based on compliance results.
5. SC-02 records transaction and product-level details.

This automated process ensures **real-time compliance evaluation** and **secure data storage**.

APPENDIX C

Trust Score Calculation Example

The trust score is dynamically updated based on farmer behavior. A sample calculation is shown below:

- Initial Trust Score: 85
- Compliant reading → +2
- Minor deviation → −3
- Chemical violation → −15

Example:

Event	Score Change	Updated Score
Compliant reading	+2	87
Compliant reading	+2	89
Minor violation	−3	86
Chemical detected	−15	71

This demonstrates how the system continuously adjusts trust based on real-time data.

APPENDIX D

System Configuration Details

The system was configured using the following setup:

- Backend: Node.js (Express API)
- Blockchain: Ethereum (Hardhat local network)
- Smart Contracts: Solidity (SC-01 to SC-04)
- Database: MongoDB
- Frontend: React.js dashboards
- Edge Simulation: Node.js/Python scripts
- Communication: MQTT protocol

This configuration enabled end-to-end simulation of the AgriTrace system.

APPENDIX E

User Roles and Permissions

The system supports three main roles:

Role	Access Type	Permissions
Farmer	Restricted	View own data, trust score
Customer	Public	Scan QR, view product trace
Admin	Full View	Monitor all data, analytics
All Roles	No Write	Cannot modify blockchain

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